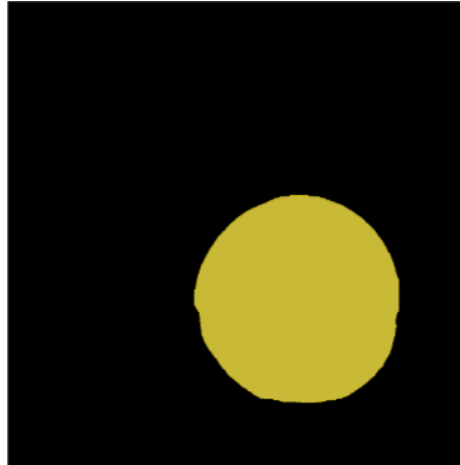
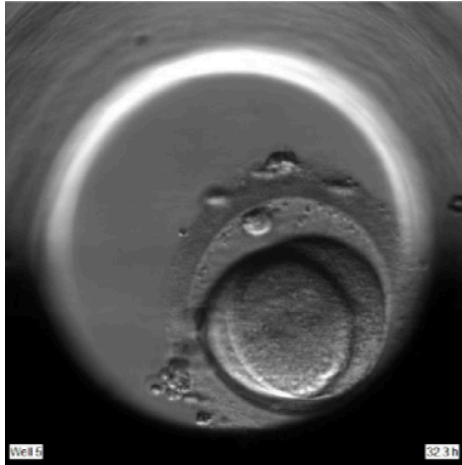


Use Case 2: Segmentation of lung cancer cells in microscopy images



Problem category: Semantic Segmentation

Recommended metrics:

- Overlap-based Metric: [Dice Similarity Coefficient \(DSC\)](#)
- Boundary-based Metric: [Normalized Surface Distance \(NSD\)](#)

Fingerprint answers:

Fingerprint	Answer
Small size of structures relative to pixel size	No
Particular importance of structure center (e.g. in cells, vessels)	No
Penalization of errors: Unequal severity of class confusions	Equal handling
Possibility of overlapping or touching target structures (e.g. medical instruments or cells)	No
Particular importance of structure boundaries	Yes
Penalization of errors: Compensation for annotation imprecisions requested	Yes

Decision between metrics:

- [Overlap-based Metric] Dice Similarity Coefficient (DSC) vs. Intersection over Union (IoU)
 - Answer: Dice Similarity Coefficient (DSC)